

Module 4: ERROR CONTROL CODING & BINARY CYCLIC CODES.

Introduction:

Various kinds of coding techniques studied in module 2, help us to achieve lower value of avg length, & thereby increasing the coding efficiency. Disadvantage is that they are variable-length codes (VLC). Due to this, a single error which occurs due to the noise present in the channel, affects more than one block-code words. Also the ^{output} data rates measured over short time-periods will fluctuate widely in VLCs.

∴ When fixed-length codes are used, a single error will affect only that block, which can be easily detected & corrected. To detect & correct errors, we go for "Error-control coding" techniques that rely on systematic addition of redundant symbols.

There are several error correcting codes & these codes are classified as Block codes & Convolutional codes. The main difference between the two are absence of memory in block codes & presence of memory in convolutional codes.

Another way of classification is "linear" & non-linear. A linear code is one in which mod-2 addition of 2 code words produces a third code word in the given block of code words.

Types of Errors:

In communication system, errors are caused by the noise present in the channel.

Random error: The transmission errors occur randomly at the receiver & hence referred as random errors.

Burst error: These errors occur due to the impulse noise or AWGN present in the channel, also they affect more than one symbol & the error is called the burst error.

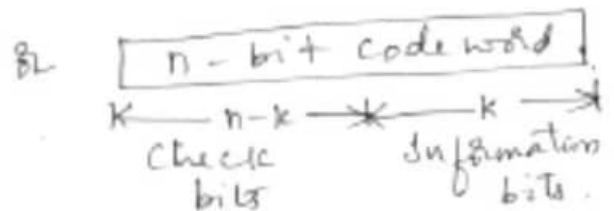
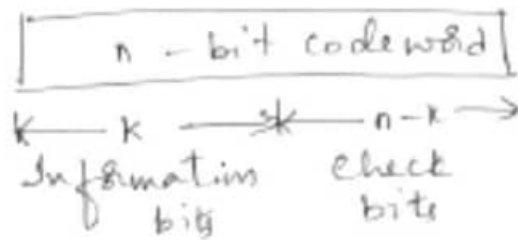
Types of codes

Error control codes are broadly categorized as Block codes & convolutional codes.

Block codes: It consists of 'k' information bits being added with $(n-k)$ redundant or check bits to form n -bit code words. At the receiver, the check bits are used to detect and correct errors which may occur in the entire n -bit code words.

A (n, k) block code is said to be linear block code (n, k) , it has to satisfy the condition that C_1 & C_2 are 2 code words of n -bits belonging to a set of (n, k) block code. then modulus-2 addition of C_1 & C_2 ($C_1 \oplus C_2$) should also be a n -bit code word belonging to the same set of (n, k) block code. A (n, k) linear block code is systematic, if k -message or information bits should appear either at the beginning or end of the code word.

eg: $C_1 \oplus C_2 = C_5$ $[C_1 \ C_2 \ C_3 \ \dots \ C_q]$
 (n, k) B.C.



Formation of linear Block codes.

Note:

modulo-2 addition	modulo-2 subtraction	modulo-2 multiplication
$0 \oplus 0 = 0$	$0 \ominus 0 = 0$	$0 \cdot 0 = 0$
$0 \oplus 1 = 1$	$0 \ominus 1 = 1$ with borrow "1"	$0 \cdot 1 = 0$
$1 \oplus 0 = 1$	$1 \ominus 0 = 1$	$1 \cdot 0 = 0$
$1 \oplus 1 = 0$	$1 \ominus 1 = 0$	$1 \cdot 1 = 1$

Matrix description of Linear Block codes.

Let the message block of k -bits (code words) be represented as a "row-vector" called message-vector given by $[D] = [d_1, d_2, \dots, d_k]$

where $d_1, d_2, d_3, \dots, d_k$ are either 0's & 1's. Thus there are 2^k distinct message-vectors.

The channel encoder systematically adds $(n-k)$ number of check-bits to form (n, k) linear block code (LBC). Then 2^k code vectors can be represented as

$$C = \{C_1, C_2, \dots, C_n\} \rightarrow \textcircled{1}$$

In a Systematic LBC, the message bits appear at the beginning of the code vector (or at the end of the code vector)

$$\therefore C_i = d_i \text{ for all } i = 1, 2, \dots, k \rightarrow (2)$$

The remaining $(n-k)$ are check bits $\therefore$ above two equations are combined as

$$[C] = \underbrace{\{C_1, C_2, \dots, C_k\}}_{k\text{-message bits}}, \underbrace{\{C_{k+1}, C_{k+2}, \dots, C_n\}}_{(n-k)\text{ check bits}} \rightarrow (3)$$

These $(n-k)$ check bits $C_{k+1}, C_{k+2}, \dots, C_n$ are derived from k -message bits using a pre-determined rule as

$$\left. \begin{aligned} C_{k+1} &= P_{11}d_1 + P_{21}d_2 + \dots + P_{k1}d_k \\ C_{k+2} &= P_{12}d_1 + P_{22}d_2 + \dots + P_{k2}d_k \\ &\vdots \\ C_n &= P_{1,n-k}d_1 + P_{2,n-k}d_2 + \dots + P_{k,n-k}d_k \end{aligned} \right\} \rightarrow (4)$$

Where $P_{11}, P_{21}, P_{12}, P_{22}, \dots$ are either 0's or 1's & the addition above performed is modulo-2 arithmetic addition

It is possible to combine the equations (2), (3) & (4) and express the result in a "matrix form" as

$$[c_1, c_2, \dots, c_k, c_{k+1}, c_{k+2}, \dots, c_n] = [d_1, d_2, \dots, d_k] \begin{bmatrix} 1 & 0 & 0 & \dots & 0 & p_{11} & p_{12} & \dots & p_{1, n-k} \\ 0 & 1 & 0 & \dots & 0 & p_{21} & p_{22} & \dots & p_{2, n-k} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 & p_{k1} & p_{k2} & \dots & p_{k, n-k} \end{bmatrix}$$

$$\underline{c} = [D][G]$$

Where $[G]$ is called as "Generator matrix" of order $(k \times n)$
 & is given by $[G] = [I_k | P]_{k \times n}$

Where I_k is unity matrix of order 'k'

$[P]$ is arbitrary matrix called "Parity Matrix"
 of order $k \times (n-k)$

& $|$ denotes the demarcation between the unit matrix I_k & parity matrix 'P'.

The Generator matrix can also be expressed as

$$[G] = [P | I_k]$$

because in systematic approach, the check bits can be placed either at the beginning or at the end of the code word & as it is a LBC same is followed.

The parity matrix $[P]$ is suitably selected to correct random & burst errors.

P1. For a Systematic (6,3) LBC, the parity matrix 'p' is given by $P = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$. Find all possible code vectors.

Solⁿ $(n, k) = (6, 3) \therefore n = 6 \text{ \& } k = 3$

We have 2^k distinct code words $\therefore 2^3 = 8$ possible code words from 000, 001, ... to 111.

The code vectors can be found using

$$[C] = [D][G]$$

$$\text{where } [G] = [I_k : P] = [I_3 : P]$$

$$\therefore [G] = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

$$\therefore [C] = [D][G] = [d_1 d_2 d_3] \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

$$[C] = [d_1 d_2 d_3 (d_1 + d_3) (d_2 + d_3) (d_1 + d_2)]$$

where the d_1, d_2, d_3 are message bits as $k = 3$, d_1, d_2, d_3

If $d_1, d_2, d_3 = 101$ then the code vector

$$[C] = [1 \ 0 \ 1 \ (1+1) \ (0+1) \ (1+0)]$$

$$[C] = [1 \ 0 \ 1 \ 0 \ 1 \ 1]$$

Thus we can find all possible code vectors for all the input message bits

Code	msg-vector	code-vector for (6,3) LBC
C_a	000	000 000
C_b	001	001 110
C_c	010	010 011
C_d	011	011 101
C_e	100	100 101
C_f	101	101 011
C_g	110	110 110
C_h	111	111 000

Parity Check matrix $[H]$ It is $(n-k) \times (n)$ matrix is used in error correction

Ex. 1, Generator matrix is given by $G = [I_k | P]$

$$\therefore [G] = [I_k | P] = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 & p_{11} & p_{12} & \dots & p_{1, n-k} \\ 0 & 1 & 0 & \dots & 0 & p_{21} & p_{22} & \dots & p_{2, n-k} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 & p_{k1} & p_{k2} & \dots & p_{k, n-k} \end{bmatrix}$$

Associated with generator matrix $[G]$, we have another matrix called Parity check matrix $[H]$, $[H] = [P^T | I_{n-k}]$

$$\therefore [H] = \begin{bmatrix} p_{11} & p_{21} & \dots & p_{k1} & 1 & 0 & 0 & \dots & 0 \\ p_{12} & p_{22} & \dots & p_{k2} & 0 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ p_{1, n-k} & p_{2, n-k} & \dots & p_{k, n-k} & 0 & 0 & 0 & \dots & 1 \end{bmatrix}$$

$\underbrace{\hspace{10em}}_{k \text{ columns}} \quad \underbrace{\hspace{10em}}_{(n-k) \text{ columns}}$

P2 If C is a valid code vector i.e., $[C] = [D][G]$, then
 Prove that $CH^T = 0$ where H^T is transpose of parity check
 matrix 'H'.

Sol:

$$\text{W.K.T, } [C] = [D][G]$$

$$[G] = [I_k; P]$$

Consider i^{th} row of $[G]$ matrix, we have

$$g_i = [0 \ 0 \ \dots \ 1 \ \dots \ 0 \ p_{i1} \ p_{i2} \ \dots \ p_{ij} \ \dots \ p_{i,n-k}]$$

$\uparrow$ i^{th} element $\uparrow$ $(k+j)^{\text{th}}$ element

Consider j^{th} row of $[H]$ matrix

$$h_j = [p_{1j} \ p_{2j} \ \dots \ p_{ij} \ \dots \ p_{kj} \ \dots \ 0 \ 0 \ \dots \ 1 \ \dots \ 0]$$

$\uparrow$ i^{th} element $\uparrow$ $(k+j)^{\text{th}}$ element

Consider $g_i \cdot h_j =$

$$[0 \ 0 \ \dots \ 1 \ \dots \ 0 \ p_{i1} \ p_{i2} \ \dots \ p_{ij} \ \dots \ p_{i,n-k}] \begin{bmatrix} p_{1j} \\ p_{2j} \\ \vdots \\ p_{ij} \\ \vdots \\ p_{kj} \\ 0 \\ 0 \\ \vdots \\ 1 \\ 0 \end{bmatrix}$$

$\uparrow$ i^{th} element $\rightarrow$ $\uparrow$ $(k+j)^{\text{th}}$ element

modulo-2 multiplication of row & column matrix

$$\begin{aligned} g_i h_j^T &= (0)(p_{1j}) + (0)(p_{2j}) + \dots + (1)(p_{ij}) + \dots + (0)(p_{kj}) + \\ &\quad (p_{i1})(0) + \dots + (p_{ij})(1) + \dots + (p_{i,n-k})(0) \\ &= p_{ij} + p_{ij} = p_{ij}(1+1) = 0. \end{aligned}$$

$$g_i h_j^T = 0$$

The above equation is true for every values of i & j
hence in the matrix form

$$[G][H^T] = 0$$

multiplying both the sides by $[D]$

$$[D][G][H^T] = 0[D]$$

$$[C][H^T] = 0 \quad \because C = DG$$

$$\underline{C \cdot H^T = 0} \quad (\text{proved})$$

Encoding circuit for (n, k) Linear Block codes:

We have code words

$$C_1 = d_1$$

$$C_2 = d_2$$

$\vdots$

$$C_k = d_k$$

$$C_{k+1} = p_{11}d_1 + p_{21}d_2 + \dots + p_{k1}d_k$$

$$C_{k+2} = p_{12}d_1 + p_{22}d_2 + \dots + p_{k2}d_k$$

$\vdots$

$$C_n = p_{1,n-k}d_1 + p_{2,n-k}d_2 + \dots + p_{k,n-k}d_k$$

The implementation of the equations in the circuit fashion results in the encoder (n, k) linear block code. The realization of encoder circuit consists of k -bit shift register, n -segment commutator & $(n-k)$ modulo 2 adders.

The entire data $d_k, d_{k-1}, \dots, d_2, d_1$ is shifted into k -bit shift register.

If no connection from d_1 to modulo 2 adder indicates that corresponding parity bit is 0 "open". If parity bit is 1, then short circuit.

PD. For a systematic $(6, 3)$ LBC, the code vector 'c' for the message input of (d_1, d_2, d_3) is given by

$[c] = [d_1 \ d_2 \ d_3 \ (d_1+d_3) \ (d_2+d_3) \ (d_1+d_2)]$. Construct the corresponding encoding circuit.

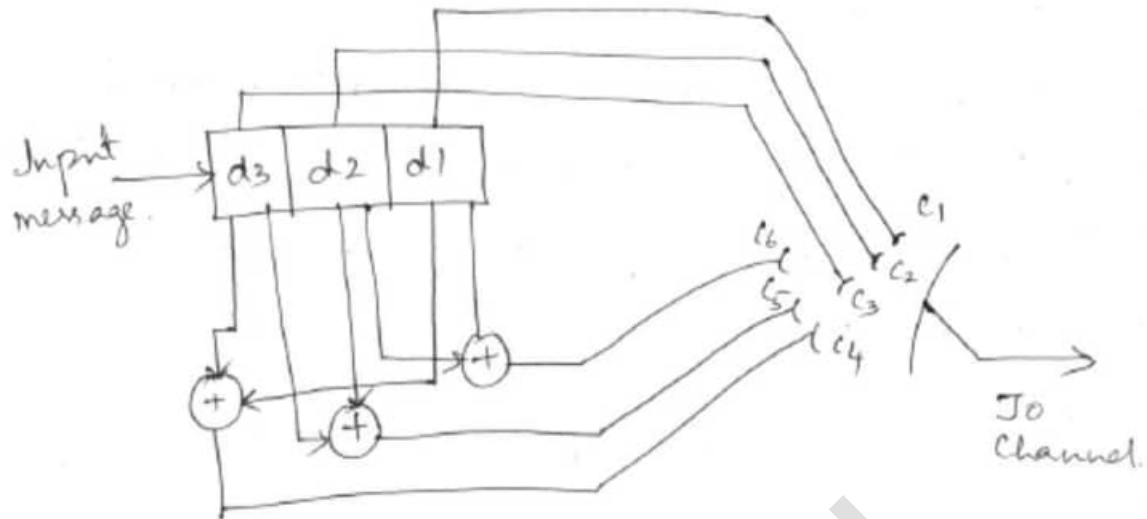
Solution: Code vector bits are given by

$$c_1 = d_1 ; c_2 = d_2 ; c_3 = d_3 ; c_4 = d_1 + d_3 ;$$

$$c_5 = d_2 + d_3 ; c_6 = d_1 + d_2.$$

Since $k=3$, we require 3-bit shift register to move the message bits into it.

$n-k = 6-3 = 3$ & hence requires 3-modulo-2 adders
 $n=6$, hence 6 segment commutators are required which in turn is connected to the channel.



Encoding circuit for (6,3) Linear Block code.

Syndrome and Error Correction.

Let us suppose that $C = (c_1 c_2 c_3 \dots c_n)$ be a valid code-vector transmitted over a noisy communication channel belonging to a (n, k) LBC.

Let $R = (r_1 r_2 \dots r_n)$ be the received vector. Due to noise in the channel $r_1 r_2 \dots r_n$ ^{may} be different from $c_1 c_2 \dots c_n$. Then the error-vector or error pattern 'E' is defined as the difference between R & C & is given by

$$E = R - C \text{ or } E = R + C$$

Since modulo 2 subtraction is same as modulo 2 addition

$\therefore$ The error vector 'E' is represented as vector by $E = (e_1 e_2 \dots e_n)$

If $E=1$ then $r_i \neq c_i$ & if $E=0$ then $r_i = c_i$. The 1's present in the error-vector 'E' represent the errors caused by noise in the channel.

If the receiver bit 'R' is known & code vector 'C' & error vector 'E' are unknown, then to find E & then C, the receiver does the decoding operation by determining $(n-k)$ vector called Syndrome vector & is defined as

$$S = R \cdot H^T = (s_1, s_2, \dots, s_{n-k})$$

Then $(n-k)$ vector S is called 'error syndrome of R'.

$$n.k.T ; R = C + E$$

$$\therefore S = (C + E) \cdot H^T$$

$$S = C \cdot H^T + E \cdot H^T$$

$$S = E \cdot H^T \quad \text{Since } C \cdot H^T = 0.$$

The receiver find E from $S = E \cdot H^T$, Since S & H^T are known. Then C can be found using $R = C + E$.

Note: ① The Syndrome S of the received vector will be zero if R is a valid Code vector.

② When $R \neq C$, then $S \neq 0$. and the receiver then detects & corrects the error.

PD. A (6,3) LBC with a parity matrix is given below. The received code vector $R = [110010]$ is given. Detect & correct the single error that has occurred due to noise.

$$P = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

Solⁿ: To detect the error the syndrome bits are to be calculated as

$$S = RH^T,$$

Received bits are given, now, we need to obtain $H^T = \begin{bmatrix} P \\ I_{n-k} \end{bmatrix}$

$$\therefore H^T = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

then the syndrome bit $S = [s_1 \ s_2 \ s_3]$ $\because n-k = 6-3 = 3$

$$\therefore S = [s_1 \ s_2 \ s_3] = RH^T = [110010] \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

By using modulo 2 multiplication & addition, syndrome bits can be obtained as,

$$S_1 = 1 + 0 + 0 + 0 + 0 + 0 = 1$$

$$S_2 = (1 \cdot 0) + (1 \cdot 1) + (0 \cdot 1) + (0 \cdot 0) + (1 \cdot 1) + (0 \cdot 0) \\ = 0 + 1 + 0 + 0 + 1 + 0$$

$$S_2 = 0$$

$$\text{Similarly } S_3 = 1 + 1 + 0 + 0 + 0 + 0$$

$$S_3 = 0$$

$$\therefore S = [S_1 \ S_2 \ S_3] = [1 \ 0 \ 0] \neq 0.$$

Hence the syndrome bits represent an error. An error has been identified, to detect and correct the H^T matrix has to be considered i.e., compare the syndrome bits with the rows of H^T . The syndrome bit $S = [1 \ 0 \ 0]$ is present in the 4th row of H^T matrix & hence 4th bit of the received vector, counting from left is having an error. The error bit has been identified, to correct that, if it is a '0' bit, it has to be replaced by '1', if it is a '1' bit, it has to be replaced by '0'. Hence the error bit can be corrected in received vector.

$\therefore$ The corrected received vector is given by $R_c = \underline{110110}$

Syndrome calculation circuit

Let the received vector $R = (r_1 r_2 \dots r_n)$
 then the Syndrome vector $S = (s_1 s_2 \dots s_{n-k}) = RH^T$

By using the modulo 2 arithmetic, we get the syndrome bits as

$$s_1 = r_1 p_{11} + r_2 p_{21} + \dots + r_k p_{k1} + r_{k+1}$$

$$s_2 = r_1 p_{12} + r_2 p_{22} + \dots + r_k p_{k2} + r_{k+2}$$

⋮

$$s_{n-k} = r_1 p_{1n-k} + r_2 p_{2n-k} + \dots + r_k p_{kn-k} + r_n$$

The above equations can be realized using circuit, which is Syndrome calculation circuit.

The received vector bits are moved into an n -bit shift register. The parity bits can be either '0' or '1'. '0' indicates open circuit & '1' indicates closed ckt. As soon as the received bits of the vector are shifted to shift register, $(n-k)$ modulo-2 adders, generate $(n-k)$ syndrome bits.

Hence we have n -bit shift register & $(n-k)$

modulo 2 adders & $(n-k)$ syndrome bits.

Knowing the Syndrome bits in a vector 'S', errors can be easily detected & corrected.

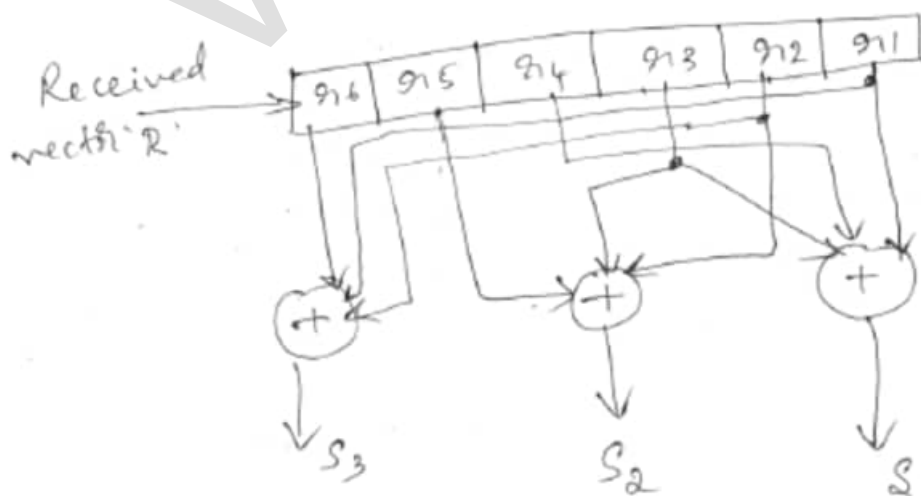
P①. For a systematic (6,3) LBC, the received vector is $R = [r_1 r_2 r_3 r_4 r_5 r_6]$, & parity matrix is given by $P = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$. Construct the corresponding Syndrome calculation circuit.

Sol: $n \times k \times T \quad S = R H^T = (S_1 S_2 S_3) \quad \because n-k=6-3=3$
 $= R \begin{bmatrix} P \\ I_{n-k} \end{bmatrix}$

$$\therefore (S_1 S_2 S_3) = [r_1 r_2 r_3 r_4 r_5 r_6] \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore [S_1 S_2 S_3] : [(r_1 + r_3 + r_4) \quad (r_2 + r_3 + r_5) \quad (r_1 + r_2 + r_6)]$$

From the above syndrome bits, Syndrome calculation circuit can be easily constructed.



Syndrome calculation circuit for (6,3) LBC.

P2. For a Systematic (7, 4) LBC, The parity matrix 'P' is given by

$$P = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

- (i) find all possible valid code-vectors
- (ii) Draw the corresponding encoding circuit
- (iii) A single error has occurred in the given received vector. Detect & correct those errors

$$R_A = 0111110 \quad R_B = 1011100$$

- (iv) Draw the syndrome calculation circuit

Sol: (i) N.K.T. $G = [I_k | P] = [I_4 | P]$

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

N.K.T, code vectors $[C] = [D][G]$

Since $k=4$, $D = [d_1 \ d_2 \ d_3 \ d_4]$

$$\therefore C = [d_1 \ d_2 \ d_3 \ d_4] \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

$$\therefore C = [c_1 \ c_2 \ c_3 \ c_4 \ c_5 \ c_6 \ c_7] = [d_1 \ d_2 \ d_3 \ d_4 \ (d_1+d_2+d_3) \ (d_1+d_2+d_4) \ (d_1+d_3+d_4)]$$

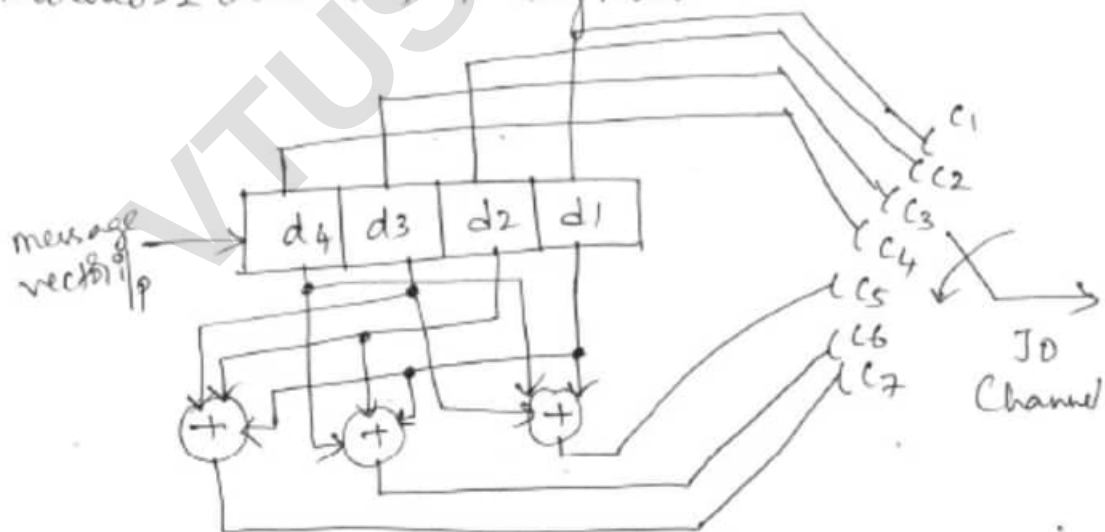
If $D = [0111]$

$C = [0111000]$

only for all $2^k = 2^4 = 16$ combination.

msg vector (D)	code vector (C)	msg vector (D)	code vector (C)
0000	0000000	1000	1000111
0001	0001011	1001	1001100
0010	0010101	1010	1010010
0011	0011110	1011	1011001
0100	0100110	1100	1100001
0101	0101101	1101	1101010
0110	0110011	1110	1110100
0111	0111000	1111	1111111

Encoding circuit requires 4-bit shift register, 3-modulo-2 adders & 7-segment commutator.



Encoding circuit for $(7,4)$ linear block code.

(iii) W.K.T $S = R + H^T$

$R_A = 0111110$

$$\therefore S_A = R_A H^T$$

$$H^T = \begin{bmatrix} P \\ -I_{n-k} \end{bmatrix}$$

$$= [0111110] \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$S_A = [s_1 s_2 s_3] = [110] \rightarrow$ Syndrome bits are present in 2nd row of $H^T \therefore$ 2nd bit of R_A from LHS is having an error, therefore corrected received vector is

$$R_{Ac} = [0011110]$$

$$R_B = [1011100]$$

$$S_B = R_B H^T$$

$$= [1011100] \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$S_B = [s_1 s_2 s_3] = [101]$$

Syndrome bits $[101]$ is present in 3rd of H^T which indicates that 3rd bit of received vector from LHS is affected by noise & is the error bit. $\therefore$ corrected received vector is $R_{Bc} = [1001100]$

Hamming weight, Hamming distance and Minimum distance of Linear Block

The concept of "weight" & "distance" between the code vectors was first introduced by R. W. Hamming.

(1) Hamming weight: Hamming weight of a code vector 'c' is defined as a non-zero bit in a code vector.

eg: If $C_1 = 0010110$ then Hamming weight is number of 1's in C_1 , i.e., $3 = H.W.$

(2) Hamming distance: Hamming distance between any two code vectors C_1 & C_2 is defined as the number of bits in which they differ.

eg: Let $C_1 = 0010110$

$C_2 = 0001100$

↓ ↓ ↓ ↓

4 bit positions are differed, hence the hamming distance between C_1 & C_2 is 4

(3) Minimum distance; (d_{min}): Minimum distance is defined as the minimum or smallest hamming distance between any two code vectors in a code, other than zero.

Q1: Consider (6,3) LBC, whose generator matrix is

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

(i) Find all code vectors
 (ii) Find all the hamming weights and distances.
 (iii) Find minimum weight parity check matrix

(iv) Draw the encoder circuit for the above codes.

Sol: For (6,3) LBC, $n=6$, $k=3$ & $n-k=6-3=3$

The code vectors can be found as $[C] = [D][G]$

$$[C] = [d_1 \ d_2 \ d_3] \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix} \quad 2^k = 2^3 = 8 \text{ distinct code vectors}$$

$$[C] = [d_1 \ d_2 \ d_3 \ (d_1+d_2) \ (d_2+d_3) \ (d_1+d_3)]$$

message vector bits	code words	Hamming weights
000	→ 000 000	→ 0
001	→ 001 011	→ 3
010	→ 010 110	→ 3
011	→ 011 101	→ 4
100	→ 100 101	→ 3
101	→ 101 110	→ 4
110	→ 110 011	→ 4
111	→ 111 000	→ 3

The hamming weights are calculated according to no. of 1's in the code vector.

The distances are given by $d_{12}=3$, $d_{13}=3$, $d_{14}=4$, $d_{23}=4$, $d_{24}=3$, $d_{34}=3$, $d_{45}=3$, $d_{56}=3$, $d_{67}=4$, $d_{78}=3$.

(iii) minimum weight parity check matrix is given as

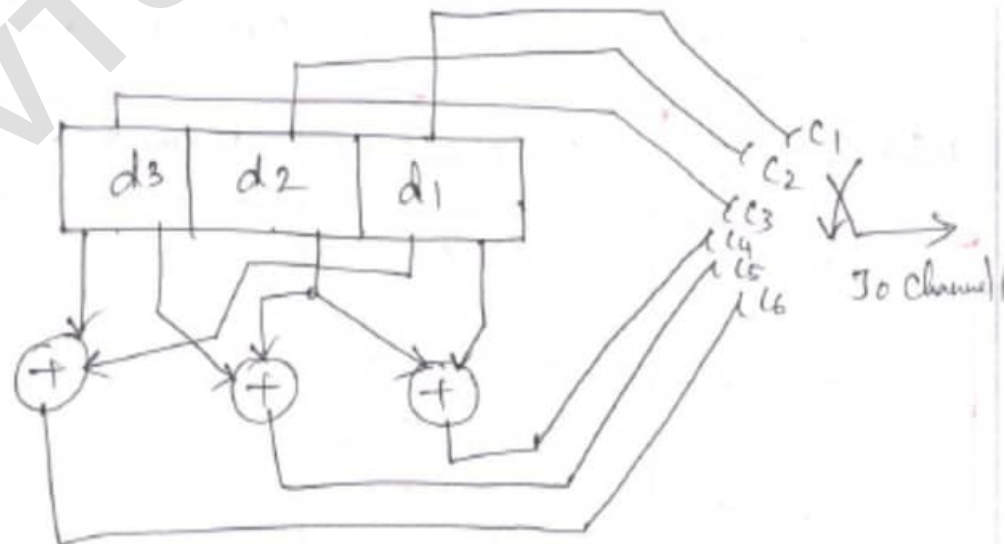
$H = [P^T \mid I_{n-k}]$ i.e., parity check matrix

$$H = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

$$P = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

$$P^T = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

(iv) Encoding circuit is given with 3-bit shift register, 3-modulo-2 adders & 6-segment commutators.



Encoding circuit (6,3) LBC.

Error Detection & Error Correction Capabilities of Linear Block codes:

Theorem 1: The minimum distance of a LBC is equal to minimum hamming weight of a non-zero code vector.

Theorem 2: Let 'C' be a (n, k) LBC with parity check matrix 'H'. For each code vector of hamming weight say 'l', there exists 'l' columns of 'H' such that the vector sum of these 'l' columns is equal to "zero vector".
i.e., "l" columns of H will add up to zero vector.

Converse of Theorem 2: Let 'C' be a (n, k) LBC with Parity Check matrix 'H'. If there exists "l" columns of 'H' whose vector sum is zero-vector, then there exists a code-vector of hamming weight "l" in C.

Corollaries of Theorem 2:

Corollary (1): Let 'C' be a LBC with parity check matrix 'H'. If no $(d-1)$ or fewer columns of H add to zero, the code has minimum hamming weight of at least 'd'.

Corollary (2): Let 'C' be a LBC with parity check matrix 'H'. The minimum hamming weight (equal to min. dist.) of 'C' is equal to the smallest number of columns of 'H' that add up to zero.

Theorem 3: A LBC with a minimum distance " $d_{\min}$ " can detect up to $(d_{\min}-1)$ errors in each code vector & can correct up to $\lfloor (d_{\min}-1)/2 \rfloor$ errors where $\lfloor (d_{\min}-1)/2 \rfloor$ denotes the largest integer no greater than $(d_{\min}-1)/2$.

The error detecting capability of a LBC with minimum distance $d_{\min}$ is $(d_{\min}-1)$

The error correcting capability of a LBC with minimum distance $d_{\min}$ is $\frac{d_{\min}-1}{2}$

Single error correcting Hamming codes:

$n \times k \times T$, Parity Check matrix is used in error detection & error correction,

$$\text{We have } H = [P^T | I_{n-k}] \text{ or } H^T = \begin{bmatrix} P \\ I_{n-k} \end{bmatrix}$$

from the above eqⁿ, it is clear that, we have $(n-k)$ number of columns. $\therefore$ each row in H^T has $(n-k)$ number of entries, which could be a "0" or "1"

Thus we have $2^{(n-k)}$ number of distinct rows.

But a row of "0"s cannot be used, as this represents the syndrome of no error. Thus we are left with $(2^{n-k}-1)$ number of distinct rows. from above H^T , we observe that, it has 'n' no. of rows (P has k rows & I_{n-k} has $(n-k)$ rows which adds up to a total of n rows)

Thus, the condition for all the rows of H^T to be distinct is that

$$2^{n-k} - 1 \geq n$$

$$\therefore 2^{n-k} \geq n+1$$

$$n-k \geq \log_2(n+1)$$

$$\underline{\underline{\text{or}}} \quad \boxed{k \leq n - \log_2(n+1)}$$

Thus, the single error correcting (n, k) hamming code has the following parameters.

code word length : $n \leq 2^{n-k} - 1$

No. of message bits : $k \leq n - \log_2(n+1)$

No. of parity check bits : $(n-k)$

Error correcting capability : $t = \frac{d_{\min}-1}{2}$

(P1). Design (n, k) hamming code, with a minimum distance of $d_{\min} = 3$ & a message length of 4-bits.

Solⁿ: Given that message length of 4 bits i.e., $k=4$ & $d_{\min} = 3$.

W.K.T. Code length $n \leq 2^{n-k} - 1$

$$n \leq 2^{n-4} - 1$$

By trial & error method, we can start for the value of n , to be greater than k .

i.e., $n=5$; $5 \leq 2^{5-4} - 1$; $5 \leq 2-1 = 1$ (not valid)

$$n=6; \quad 6 \leq 2^{6-4}-1; \quad 6 \leq 2^2-1 = 3 \text{ (not valid)}$$

$$n=7; \quad 7 \leq 2^{7-4}-1; \quad 7 \leq 8-1 = 7 \text{ (valid)}$$

$\therefore$ we have $(7, 4)$ LBC, is also called Hamming code.

$$H^T = \begin{bmatrix} P \\ I_{n-k} \end{bmatrix} = \begin{bmatrix} P \\ I_3 \end{bmatrix}$$

$$H^T = \begin{bmatrix} [P] \\ 100 \\ 010 \\ 001 \end{bmatrix}$$

Parity matrix, $[P]$ has $k=4$ rows & 3 columns. ^{$(n-k)$}
has to be suitably chosen. The requirements for choosing the $[P]$ matrix are.

- 1) H^T should not contain a row of 0's as this represents the syndrome of no error.
- 2) No two rows of H^T must be same.

Here, all the 7 rows of H^T must be distinct.

There are totally $2^{n-k} = 2^{7-4} = 2^3$ combinations of 3 bit numbers from 000, 001, ... to 111.

Out of these 8 combinations, 000 cannot be used as it indicates syndrome of no error & also 001, 010 & 100 cannot be used as they are the elements of identity matrix. $I_{n-k} = I_3$. Thus, we are left with only four combinations, 011, 101, 110 & 111, to be chosen as the 4 rows of $[P]$.

Thus, these 4 combinations can be arranged in $4! = 24$ ways in the rows of $[P]$. Any of these combinations can be used in H^T , which can be used for error correction.

Let us choose a parity matrix $P = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

$$\therefore H^T = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ \hline 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\therefore$ The parity check matrix 'H' is also called "Hamming matrix" given by

$$H = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

It can be observed no two columns of H add up to zero, but three columns (1^{st} , 4^{th} & 5^{th}) add up to zero-vector so that $d_{min} = 3$. which is the condition given in problem.

The generator matrix $[G]$ is given by $[G] = [I_k | P]$

$$[G] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}$$

The various code vectors can be found using

$$[C] = [D][G]$$

$$[C] = [d_1 d_2 d_3 d_4] [G]$$

$$[C] = [d_1 d_2 d_3 d_4 (d_2 + d_3 + d_4) (d_1 + d_3 + d_4) (d_1 + d_2 + d_4)]$$

$\therefore$ If $D = [1010]$ then $[C] = [1010101]$

Many other code vectors can be found.

Error correction:

Since $d_{\min} = 3$, the error correcting capability of the code is $t = \frac{d_{\min} - 1}{2} = \frac{3 - 1}{2} = 1$.

$\therefore (7, 4)$ hamming code so formed is a SEC code.

Q2) Design a single error correcting code with a message block size of 11 & show that by an example that it can correct single error.

Sol: Single error correcting hamming code can be designed by considering the code length 2^n , which must satisfy the condition, $n \leq 2^{n-k} - 1$

given; $k = 1$, $n \leq 2^{n-1} - 1$

By trial & error, least integer value of n to be selected greater than 11 i.e.,

$$n = 12; \quad 12 \leq 2^{12-1} - 1 = 1 \quad (\text{not valid})$$

$$n=13; \quad 13 \leq 2^{13-11} - 1 = 3 \quad (\text{not valid})$$

$$n=14; \quad 14 \leq 2^{14-11} - 1 = 7 \quad (\text{not valid})$$

$$n=15; \quad 15 \leq 2^{15-11} - 1 = 15 \quad (\text{valid})$$

$\therefore (15, 11)$ hamming code & we have

$$H^T = \begin{bmatrix} P_{k \times (n-k)} \\ I_{n-k} \end{bmatrix} = \begin{bmatrix} P_{11 \times 4} \\ I_4 \end{bmatrix} =$$

The parity matrix can be written without 0000, 1000, 0100, 0010, 1000 i.e., out of $2^{15-11} = 2^4 = 16$ combⁿ.
 $16 - 5 = 11$, write all 11 elements in the parity matrix

$$\therefore H^T = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ \hline 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Then the $[H]$ matrix is given by $H = [P^T | I_{n-k}]$

$$H = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Table lookup decoding (Syndrome Decoding) Using the standard array.

All the error patterns that differ at most by a code-vector have the same syndrome i.e., each co-set is characterized by an unique syndrome.

Since the received vector 'r' may be any of the n-tuples (i.e., 2^n combinations). Using the eqⁿ E14¹, the received code-vectors may be partitioned into 2^k disjoint co-sets and try to identify the received vector 'r' in them. This is done by preparing "Standard array".

P1) Construct the standard array for the (6,3). The code vector is (000000) (001110) (010011) (011101) (100101) (101011) (110110) & (111000) $c = [d_1 \ d_2 \ d_3 \ (d_1+d_2) \ (d_2+d_3) \ (d_1+d_3)]$

Syndrome	Co-set leader							
000	000000	001110	010011	011101	100101	101011	110110	111000
101	100000	101110	110011	111101	000101	001011	010110	011000
011	010000	011110	000011	001101	110101	111011	100110	101000
110	001000	000110	011011	010101	101001	100011	111110	110000
100	000100	001010	010111	011001	100001	101111	110010	111100
010	000010	001100	010001	011111	100110	101001	110100	111010
001	000001	001111	010010	011100	100100	101010	110111	111001
111	100010	101110	110001	111111	000111	001001	010100	011010

Standard array for (6,3) LBC

The Standard array is formed in the following way.

1) In the Syndrome column, 1st row, place the Syndrome of no error i.e., all zero elements. The other elements of the column should be the elements of H^T ; as there should be 8 rows in the column, but from H^T & error bits, we have 7 elements (rows) $\therefore$ left over combination is 111, which can be placed in the last row of Syndrome.

2) The first row of the table are the code vectors given in the question.

3) The 2nd column in the table is co-set leader, which can be formed by considering a single bit error separately, considering the individual bits with error. For the 2nd and subsequent rows of the co-set leader. As the LBC is (6,3), we are left with last row. $\therefore$ Any two bits with error should be considered.

4) The other elements of the table is filled by considering the first row except Syndrome & co-set leader. Then by considering the co-set leader column, the 2nd & subsequent rows ^{of the table} are filled by replacing the error bit of co-set leader (i.e., if error bit is 1 in the ~~co-set~~ code-vector, then replace it by zero & vice-versa). Continue & fill the table.

Decoding using Standard array table

Syndrome $S = R H^T$ Assume $R = 100100$

$$S = [100100] \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$S = [s_1 \ s_2 \ s_3] = [0 \ 0 \ 1]$$

which is present in 6th row of H^T , which indicates that, error is present in the 6th ~~row~~ bit of the received vector which is zero in the given received vector, which should be changed to one to correct it $\therefore R_c = 100101$, $E = 000001$.

P2. The parity check bits of a (8, 4) block code are generated by

$$\begin{aligned} C_5 &= d_1 + d_2 + d_4 \\ C_6 &= d_1 + d_2 + d_3 \\ C_7 &= d_1 + d_3 + d_4 \\ C_8 &= d_2 + d_3 + d_4 \end{aligned}$$

where d_1, d_2, d_3 & d_4 are message bits

(i) Find the generator matrix & parity check matrix for this code.

(ii) Find the minimum weight of this code

(iii) Show that it is capable of correcting all single error patterns & capable of detecting all error & prepare Syndrome Table.

(i) The code vector for $(8, 4)$ LBC can be written as

$$[C] = [D][G]$$

$$= [d_1 \ d_2 \ d_3 \ d_4 \ (d_1+d_2+d_4) \ (d_1+d_2+d_3) \ (d_1+d_3+d_4) \ (d_2+d_3+d_4)]$$

$$= [d_1 \ d_2 \ d_3 \ d_4] \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 1 \end{bmatrix}$$

$\therefore$ The generator matrix is

$$[G] = [I_4 | P] = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 1 \end{bmatrix}$$

parity check matrix is given by $[H] = [P^T | I_{n-k}]$

$$H = \begin{bmatrix} 1 & 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

(ii) To find the minimum weight of the code, find all possible $2^k = 2^4 = 16$ code vectors using $C = [D][G]$. then the minimum distance of the code vector is, minimum hamming weight other than zero code vector is minimum weight.

msg vector	code vector	Hw	msg vector	code vector	Hw
0000	00000000	0	1000	10001110	4
0001	00011011	4	1001	10010101	4
0010	00100111	4	1010	10101001	4
0011	00111100	4	1011	10110010	4
0100	01001101	4	1100	11000011	4
0101	01010110	4	1101	11011001	5
0110	01101010	4	1110	11100101	5
0111	01110001	4	1111	11111111	8

It can be observed that minimum weight = 4

(iii) Let us construct the Syndrome table showing single error patterns & corresponding Syndrome.

Syndrome bit = $n - k = 8 - 4 = 4$ bits.

Single error pattern. Co-set leader	Syndrome
10000000	1110
01000000	1101
00100000	0111
00010000	1011
00001000	1000
00000100	0100
00000010	0010
00000001	0001

#^T matrix elements

Syndrome is calculated as $S = R H^T$

Assume any one bit with error.

Consider $R = 10000000$.

$$\therefore S = RH^T$$

$$S = [s_1 \ s_2 \ s_3 \ s_4] = [10000000] \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$S = [1110]$$

Which is present in 1st row of H^T . $\therefore$ 1st bit is in error $\therefore$ corrected received vector is

$$R_c = 00000000$$

Binary cyclic codes:

It is a subclass of linear block codes. The cyclic codes have a mathematical structure & algebraic structures, we can develop a polynomial representation for cyclic codes & can be used in encoding & decoding.

Definition: A (n, k) LBC is said to be a cyclic code, if every cyclic shifts of the code is also a code vector of "C".

If $V = (v_0, v_1, v_2, \dots, v_{n-1})$
The polynomial representation can be written as

$$V(x) = v_0 + v_1x + v_2x^2 + \dots + v_{n-1}x^{n-1}$$

Generator & Parity check matrices of (7,4) cyclic codes.

The generator polynomial $g(x)$ & parity check polynomial $h(x)$ defined by $G = [P | I_k]$

for (7,4) LBC,

$$g(x) = 1 + x + x^3 \quad \text{as } n=7,$$

$$g(x) = (1) + (1)(x) + (0)(x^2) + (1)(x^3) + (0)(x^4) + (0)(x^5) + (0)(x^6)$$

$$g = 1101000$$

$$x \cdot g(x) = x(1 + x + x^3) = x + x^2 + x^4$$

$$g = 0110100$$

$$x^2 g(x) = x^2(1 + x + x^3) = x^2 + x^3 + x^5$$

$$g = 0011010$$

$$x^3 g(x) = x^3(1 + x + x^3) = x^3 + x^4 + x^6$$

$$g = 0001101$$

$$G = \begin{bmatrix} 1101000 \\ 0110100 \\ 0011010 \\ 0001101 \end{bmatrix}$$

but it is not in systematic form. Systematic form

is obtained by adding 1st & 3rd row and placing the result in 3rd row, & also add 1st, 2nd & 4th are added & placed in 1st row.

$$G = \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 1 \end{array} \right] = [P | I_k]$$

$$\text{w.k.t } g(x) \cdot h(x) = x^n + 1$$

$$\text{for } (7,4); n=7 \therefore x^7 + 1 = g(x) \cdot h(x)$$

$$\therefore \text{parity check polynomial } h(x) = \frac{x^7+1}{g(x)}$$

$$h(x) = \frac{x^7+1}{x^3+x+1}$$

$$\begin{array}{r} x^3+x+1 \overline{) x^7+1} \quad (x^4+x^2+x+1) \\ \underline{x^7+x^5+x^4} \\ x^5+x^4+1 \\ \underline{x^5+x^3+x^2} \\ x^4+x^3+x^2+1 \\ \underline{x^4+x^2+x} \\ x^3+x+1 \\ \underline{x^3+x+1} \\ 0 \end{array}$$

$\therefore$ The parity check polynomial $h(x) = 1+x+x^2+x^4$.
Reciprocal of $h(x)$ is defined as $x^k h(x^{-1})$. This polynomial is also a factor of $1+x^n$.

$x^4 h(x^{-1})$ for $(7, 4)$ cyclic code is

$$h(x) = 1+x+x^2+x^4$$

$$h(x^{-1}) = 1 + \frac{1}{x} + \frac{1}{x^2} + \frac{1}{x^4}$$

$$x^4 h(x^{-1}) = x^4 + x^3 + x^2 + 1 \rightarrow 1011100$$

$$x^5 h(x^{-1}) = x^5 + x^4 + x^3 + x \rightarrow 0101110$$

$$x^6 h(x^{-1}) = x^6 + x^5 + x^4 + x^2 \rightarrow 0010111$$

$$H = \begin{bmatrix} 1 & 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \end{bmatrix}$$

But is not in std., for $[I_{n-k} \mid P]$

$\therefore$ 1st & 3rd row added & placed in 1st row.

$$\therefore H = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \end{bmatrix}$$

Encoding circuit using $(n-k)$ bit Shift register

$[R] \rightarrow$ Flipflops that make up a shift register

$(+)$ $\rightarrow$ modulo 2 adders

$g_i = 0$ if open ckt & $g_i = 1$ if closed path

$[gate] \rightarrow$ And gate.

Syndrome calculation - Error detection & correction.

(n, k) Syndrome calculation ckt for (n, k) cyclic code

